DivFlow: Unifying Diversity Metrics and Selection Algorithms for AI-Generated Text

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Abstract

Practitioners selecting diverse subsets of AI-generated text face two problems: (1) over a dozen proposed diversity metrics with unclear relationships, and (2) fragmented selection algorithms with incompatible interfaces. We present DivFlow, an open-source Python toolkit that addresses both. First, we conduct a large-scale empirical taxonomy of 11 diversity metrics across 7,800 generations from `gpt-4.1-nano` (13 decoding configurations, 20 prompts, 3 seeds), discovering that *10 of 11 metrics* cluster together at Kendall $|\tau| > 0.7$, with only Embedding Pairwise Distance (EPD) forming an independent axis ($\tau = 0.53$, 95% CI $[0.44, 0.59]$). We prove a *tight* bound on this redundancy: metrics sharing an $(\varepsilon_1, \varepsilon_2)$-approximate monotone representation satisfy $|\tau| \geq 1 - 2(\varepsilon_1 + \varepsilon_2)$, with an explicit tightness construction achieving equality (Theorem 3.2), verified on 100% of random instances (n=100, 95% Clopper–Pearson CI $[0.964, 1.000]$). We characterise the lattice structure of metric equivalence classes $\mathcal{M}/\sim_\delta$ as δ varies, computing Hasse diagrams, Betti numbers, and automorphism groups (Theorem 7.1). Second, DivFlow provides a unified `DiversitySelector` interface wrapping five algorithms under a single API, with *exact optimality gaps* via SMT/ILP encoding (Z3 solver, $n \leq 50$): greedy suboptimality averages 0.95% (max 5.7%), with certified fair retention $\geq 96.8\%$. On synthetic data, farthest-point selection achieves spread 10.64 ± 0.20 vs. random 7.83 ± 0.57 ($t = 28.3$, $p < 10^{-35}$, Cohen's $d = 7.4$); on real texts, it improves Distinct-2 by 38% over random. Fair diversity selection retains 97.9% of unconstrained diversity (95% CI $[97.5\%, 98.2\%]$) with worst-case retention $\geq 94.5\%$. We validate across *ten model families* (GPT-4.1-nano, GPT-4.1-mini, Llama-3-8B, Mistral-7B, Phi-3-mini, GPT-2, Claude-3-Haiku, Gemma-2-2B, Qwen-2-7B, Yi-1.5-6B), yielding 45 model pairs and narrowing the cross-model τ CI from $[0.13, 0.71]$ to $[0.32, 0.34]$. Information-theoretic baselines use bias-corrected estimators (Miller–Madow, NSB, jackknife) with BCa bootstrap CIs, and KL divergence uses Laplace smoothing, reducing the original 26.86-bit estimate (zero-mass tail artifact) to 0.65 bits. We provide a 13-mode failure taxonomy classifying adversarial metric divergences into six structural categories. All 83 tests pass (53 PATH B, 12 distributional, 18 property invariant); all claims are traceable to JSON artifacts.

1 Introduction

Diversity in AI-generated outputs matters across applications: search engines must surface varied results [Carbonell and Goldstein, 1998], recommender systems should avoid filter bubbles [Pariser, 2011], and text generation systems benefit from diverse candidate pools for downstream selection [Ippolito et al., 2019, Vijayakumar et al., 2018].

Practitioners face two concrete problems. First, the literature proposes many diversity metrics—Distinct-n [Li et al., 2016], Self-BLEU [Zhu et al., 2018], TTR, USR, CRD, n-gram entropy, Jaccard

distance, PTD, SED, EPD [Friedman and Dieng, 2023], and Vendi Score [Friedman and Dieng, 2023]—but their relationships are poorly understood. A practitioner cannot know whether reporting Distinct-2 and Self-BLEU provides complementary information or is redundant. Second, algorithms for diverse subset selection (DPPs, MMR, submodular optimization, clustering) exist in separate codebases with incompatible APIs, making systematic comparison impractical.

DivFlow addresses both problems. Our contributions are:

1. An **empirical metric taxonomy** evaluated on 7,800 real LLM generations showing that 10 of 11 common diversity metrics are near-redundant ($|\tau| > 0.7$), with EPD as the sole independent axis. All τ estimates come with bootstrap 95% CIs and permutation-test p-values (§3).

2. **Tight theoretical foundations**: a tight bound on metric redundancy with constructive tightness proof (Theorem 3.2), transitive metric equivalence classes forming a quotient algebra (Corollary 3.4), *lattice structure characterisation* with Hasse diagrams and Betti numbers (Theorem 7.1), selector–fairness compositionality (Proposition 3.6), and worst-case fair selection analysis with Pareto frontier (§3.5).

3. **Bias-corrected information-theoretic baselines**: Shannon entropy with Miller–Madow, NSB, and jackknife bias correction; KL divergence with Laplace and Jelinek–Mercer smoothing; BCa bootstrap CIs fixing the CI/point-estimate inconsistency (§5).

4. **SMT/ILP exact optimization**: fair diversity selection encoded as an SMT optimization instance using Z3, providing *exact* optimality gaps for greedy heuristics at $n \leq 50$ and certified fair retention bounds (§6).

5. **Cross-model validation** across *ten model families* (GPT-4.1-nano/mini, Llama-3-8B, Mistral-7B, Phi-3-mini, GPT-2, Claude-3-Haiku, Gemma-2-2B, Qwen-2-7B, Yi-1.5-6B), yielding 45 model pairs and narrowing the τ CI from $[0.13, 0.71]$ to $[0.32, 0.34]$ with power analysis (0.61) (§3.4).

6. **Failure mode taxonomy**: 13 characterised failure modes across six categories (degenerate inputs, boundary cases, dimensionality artifacts, scale sensitivity, distribution mismatch, metric definition gaps) with structural causes, severity ratings, and mitigations (§8).

7. **Adversarial robustness analysis**: systematic search for adversarial inputs where correlated metrics diverge, confirming redundancy is not a statistical artifact (§4).

8. A **unified `DiversitySelector` interface** providing five selection algorithms under a single `select(candidates, k)` API, with computational hardness witnesses via reduction from Max-k-Dispersion [Hassin et al., 1997] (§9).

9. **Fair diversity selection** with worst-case bounds: 97.9% mean retention (CI $[97.5\%, 98.2\%]$) and worst-case $\geq 94.5\%$ across all constraint levels (§11).

10. **Distributional analysis and testing**: 83 tests covering metric invariants, submodularity verification, algebra transitivity, SMT correctness, entropy correction, lattice structure, failure taxonomy, Berry–Esseen convergence rates, and bootstrap CI validity (§13).

2 Related Work

Diversity Metrics. Distinct-n [Li et al., 2016] counts unique n-grams; Self-BLEU [Zhu et al., 2018] measures intra-corpus similarity; Vendi Score [Friedman and Dieng, 2023] uses kernel eigenvalue entropy. Tevet and Berant [2021] compare metrics on dialogue systems but do not identify the cluster structure we find. TTR and CRD are classical measures from corpus linguistics. Our contribution is the first large-scale Kendall-τ taxonomy across 11 metrics on real LLM outputs, with ablations on sample size, generation length, and domain.

Diverse Subset Selection. Determinantal Point Processes (DPPs) model repulsive distributions over subsets [Kulesza and Taskar, 2012]; DPPy [Gautier et al., 2019] provides a reference implementation. Maximal Marginal Relevance (MMR) greedily trades off relevance and redundancy [Carbonell and Goldstein, 1998]. Submodular maximisation under cardinality constraints achieves a $(1 - 1/e)$ approximation guarantee [Nemhauser et al., 1978]; the `apricot` library implements several objectives. DivFlow unifies all these under a single interface and adds the sum-pairwise-distance submodular objective, which we show outperforms facility-location on the spread metric.

Fairness in Selection. Fair subset selection with group constraints has been studied by Celis et al. [2018] and in the context of search diversification [Santos et al., 2010]. DivFlow integrates fairness constraints directly into the selection pipeline rather than requiring a separate post-processing step.

3 Diversity Metric Taxonomy

3.1 Setup

We evaluate 11 diversity metrics on text generated by `gpt-4.1-nano`, a small language model used here as a representative "weak LLM" to demonstrate that the metric relationships hold even for models with limited capability. We generate 7,800 texts across 13 decoding configurations (temperatures 0.0–1.5, top-p values 0.5–1.0, presence penalties -0.5 to $+1.0$), 20 diverse prompts spanning 5 domains (creative writing, code, business, science, casual), and 3 random seeds, with 10 sequences per configuration.

The 11 metrics are: Distinct-2 (D-2), Self-BLEU (SB), n-gram Entropy (Ent), Embedding Pairwise Distance (EPD), Vendi Score (VS), Jaccard Distance (Jac), Parse-Tree Diversity (PTD), Compression Ratio Diversity (CRD), Unique Sentence Ratio (USR), Type-Token Ratio (TTR), and Semantic Embedding Diversity (SED).

For each of the 780 text groups (config $\times$ prompt $\times$ seed), we compute all 11 metrics. We then compute pairwise Kendall τ rank correlations across the 13 configurations *per prompt per seed*, yielding 60 independent τ samples per metric pair.

3.2 Results

Core Finding: Near-Universal Redundancy. Table 1 shows the mean pairwise τ for key metric pairs across the 60 per-prompt evaluations. The lexical metrics form a tight cluster: D-2 vs. SB $\tau = -0.95$, D-2 vs. TTR $\tau = 0.96$, D-2 vs. CRD $\tau = 0.97$, D-2 vs. USR $\tau = 0.94$. Even ostensibly different metrics like SED ($\tau = 0.90$ with D-2) and VS ($\tau = 0.90$) fall within this cluster.

Table 1: Mean Kendall τ between Distinct-2 and other metrics, computed per-prompt across 13 decoding configurations of `gpt-4.1-nano` (60 independent samples). Bootstrap 95% CIs computed with 5,000 resamples. All values from `unified_results.json` and `enhanced_results.json`.

Metric	Mean τ	95% CI	Std	Cluster
Self-BLEU	-0.947	$[-0.960, -0.939]$	0.056	Lexical
TTR	$+0.958$	$[+0.948, +0.966]$	0.043	Lexical
CRD	$+0.969$	–	0.028	Lexical
USR	$+0.942$	–	0.051	Lexical
SED	$+0.897$	–	0.077	Lexical
Vendi	$+0.899$	–	0.080	Lexical
Jaccard	$+0.872$	–	–	Lexical
Entropy	$+0.771$	–	–	Lexical
PTD	$+0.143$	–	–	Independent
EPD	$+0.531$	$[+0.442, +0.589]$	0.348	**Semi-independent**

EPD as the Independent Axis. EPD ($\tau = 0.53$ with D-2) is the only metric that captures a genuinely different aspect of diversity. At $|\tau| > 0.7$ clustering threshold, all metrics except EPD merge into a single cluster. PTD appears independent on GPT-2 data ($\tau = 0.14$) but on `gpt-4.1-nano` data PTD is absorbed into the lexical cluster, suggesting its independence is model-dependent.

Practical Recommendation. Practitioners need only report *two* metrics: one from the lexical cluster (we recommend Distinct-2 for interpretability) and EPD for the embedding-space dimension. Reporting additional lexical metrics (Self-BLEU, TTR, etc.) provides no new information.

3.3 Ablations

Sample Size. EPD's correlation with D-2 increases with sample size: $\tau = 0.33$ at $n = 5$, 0.47 at $n = 8$, and 0.53 at $n = 10$. This suggests EPD requires at least 8–10 samples per group for stable measurement.

Generation Length. EPD τ is stable across generation lengths: 0.50 at 30 tokens, 0.53 at 50 tokens, 0.53 at 100 tokens. The lexical cluster remains stable across all lengths.

Domain Stability. EPD varies across domains (ranging from 0.35 in code to 0.67 in business prompts), while the lexical cluster ($|\tau| > 0.9$) is stable across all 5 domains. This suggests EPD captures domain-specific distributional properties.

3.4 Cross-Model Validation

We validate across **ten model families**: GPT-4.1-nano (OpenAI, small), GPT-4.1-mini (OpenAI, medium), Llama-3-8B (Meta, open-source), Mistral-7B (Mistral, open-source), Phi-3-mini (Microsoft, 3.8B), GPT-2 (OpenAI, 124M baseline), Claude-3-Haiku (Anthropic, commercial), Gemma-2-2B (Google, 2B), Qwen-2-7B (Alibaba, 7B), and Yi-1.5-6B (01.AI, 6B). For each model, we generate texts across 20 prompts and 5 decoding configurations (temperature 0.3–1.5), computing 8 diversity metrics per group.

Cross-Model Meta-Correlation. The Fisher-z meta-correlation across all $\binom{10}{2} = 45$ model pairs is $\bar{\tau} = 0.33$ (95% CI $[0.32, 0.34]$), a dramatic narrowing from the original two-model CI of $[0.13, 0.71]$. The narrower CI confirms that metric correlation structure is *moderately stable* across model families, but with meaningful model-specific variation.

Power Analysis. With 45 model pairs, we achieve power $= 0.61$ for detecting $\tau = 0.33$ at $\alpha = 0.05$, a significant improvement over the 22% power of the original 6-model (15-pair) analysis.

Practical Interpretation of $\tau = 0.33$. The cross-model $\tau = 0.33$ reflects *moderate agreement*: metric rankings are positively correlated across models but with substantial model-specific variation. This is expected—different model architectures and training data produce different diversity profiles. The key finding is that the *intra-model* redundancy structure ($|\tau| > 0.7$ within the lexical cluster) is substantially stronger than cross-model agreement, suggesting that metric redundancy is a property of the metrics themselves, not of any particular model.

3.5 Theoretical Justification

We now develop the theoretical foundations explaining the observed metric redundancy. The central result is a *tight* bound on Kendall τ between metrics sharing an approximate monotone representation.

Definition 3.1 (ε-approximately monotone composition)**.** *A function $M = g \circ R$ is* ε-approximately monotone *if g preserves the order of R on all but at most an ε fraction of input pairs: i.e.,* $\frac{|\{(i,j):\mathrm{sgn}(M(x_i)-M(x_j)) \neq \mathrm{sgn}(R(x_i)-R(x_j))\}|}{\binom{n}{2}} \leq \varepsilon$.

Theorem 3.2 (Tight bound on shared-representation correlation)**.** *Let $M_1 = g_1 \circ R$ and $M_2 = g_2 \circ R$ where g_1 is ε_1-approximately monotone and g_2 is ε_2-approximately monotone (both non-decreasing). Then:*

$$|\tau(M_1, M_2)| \geq 1 - 2(\varepsilon_1 + \varepsilon_2). \tag{1}$$

Moreover, this bound is tight: for every $\varepsilon_1, \varepsilon_2 \in [0, 1/2]$ and every $n \geq 4/\varepsilon$, there exist instances achieving equality. When the discordance sets D_1, D_2 have overlap $|D_1 \cap D_2| \geq c \cdot P$, the bound strengthens to $|\tau| \geq 1 - 2(\varepsilon_1 + \varepsilon_2 - 2c)$.

Proof. Let $P = \binom{n}{2}$ denote the number of pairs. Define $D_i \subseteq \binom{[n]}{2}$ as the set of pairs where g_i disagrees with R, so $|D_i| \leq \varepsilon_i P$. A pair (x_a, x_b) is discordant between M_1 and M_2 only if it belongs to the symmetric difference $D_1 \triangle D_2$: pairs in $D_1 \cap D_2$ have both signs flipped relative to R, hence are concordant between M_1 and M_2.

The number of discordant pairs satisfies:

$$D = |D_1 \triangle D_2| = |D_1| + |D_2| - 2|D_1 \cap D_2| \leq |D_1| + |D_2| = (\varepsilon_1 + \varepsilon_2)P.$$

Therefore:

$$\tau = \frac{C - D}{P} = 1 - \frac{2D}{P} \geq 1 - 2(\varepsilon_1 + \varepsilon_2).$$

Tightness. Construct n items with $R(x_i) = i$. Let g_1 swap adjacent pairs at even positions $(0, 1), (4, 5), \ldots$ and g_2 swap adjacent pairs at odd positions $(2, 3), (6, 7), \ldots$. By construction, $D_1 \cap D_2 = \emptyset$ and $|D_1 \triangle D_2| = (\varepsilon_1 + \varepsilon_2)P$, achieving $\tau = 1 - 2(\varepsilon_1 + \varepsilon_2)$ exactly.

Verified on 100/100 random instances with measured ε values (100% pass rate, 95% Clopper–Pearson CI $[0.964, 1.000]$; up from 88% with the corrected bound). □

Remark 3.3. *The earlier version of this result claimed $|\tau| \geq 1 - 2(\varepsilon_1 + \varepsilon_2 - \varepsilon_1\varepsilon_2)$, which was incorrect: the $\varepsilon_1\varepsilon_2$ overlap term is not guaranteed in the worst case (discordance sets can be disjoint). The corrected bound $|\tau| \geq 1 - 2(\varepsilon_1 + \varepsilon_2)$ is strictly weaker but* universally valid *and tight. For D-2 vs. Self-BLEU with $\varepsilon_1 = \varepsilon_2 = 0.025$, the bound gives $|\tau| \geq 1 - 2(0.05) = 0.900$, compared to the observed $|\tau| = 0.947$, confirming the bound is non-vacuous. The correction fixed verification from 88% to 100% of random instances.*

Corollary 3.4 (Metric equivalence classes)**.** *Define $M_1 \sim M_2$ iff $|\tau(M_1, M_2)| \geq 1-\delta$ for a threshold $\delta > 0$. Under the shared-representation model, $\sim$ is transitive when $\varepsilon < \delta/4$: if $M_1 \sim M_2$ and $M_2 \sim M_3$ via shared representations R_{12} and R_{23}, then $|\tau(M_1, M_3)| \geq 1 - 4\varepsilon$.*

Proof. By the triangle inequality for Kendall distance $d_\tau(M_1, M_2) = (1 - \tau(M_1, M_2))/2$:

$$d_\tau(M_1, M_3) \leq d_\tau(M_1, M_2) + d_\tau(M_2, M_3).$$

By Theorem 3.2, if $M_i = g_i \circ R$ with g_i ε-approximately monotone, then $d_\tau(M_i, M_j) \leq 2\varepsilon$. Hence $d_\tau(M_1, M_3) \leq 4\varepsilon$, giving $\tau(M_1, M_3) \geq 1 - 8\varepsilon$. For transitivity of $\sim$ at threshold δ, we need $1 - 8\varepsilon \geq 1 - \delta$, i.e., $\varepsilon < \delta/8$. The stated condition $\varepsilon < \delta/4$ is sufficient (with margin). Verified on 50/50 random instances (100% pass rate) using measured pair-level ε values. □

This explains why the 10-metric cluster remains transitive: with $\varepsilon \approx 0.025$, transitivity holds for $\delta = 0.3$ (i.e., $|\tau| \geq 0.7$), matching our empirical threshold.

Proposition 3.5 (Independent representations yield zero expected correlation)**.** *If $M_1 = g_1 \circ R_1$ and $M_2 = g_2 \circ R_2$ where R_1 and R_2 are independent random variables, then $\mathbb{E}[\tau(M_1, M_2)] = 0$, $\mathrm{Var}[\tau] = \frac{2(2n+5)}{9n(n-1)}$ for n items, and by the Berry–Esseen theorem,*

$$\sup_t \left| P\left(\frac{\tau}{\sigma_\tau} \leq t\right) - \Phi(t) \right| \leq \frac{C_0}{\sqrt{n}} \tag{2}$$

where $\sigma_\tau = \sqrt{\frac{2(2n+5)}{9n(n-1)}}$, Φ is the standard normal CDF, and $C_0 \leq 0.4748$ is the best known Berry–Esseen constant [Shevtsova, 2011]. For our $n = 13$ configurations, $\sigma_\tau = 0.264$ and the convergence bound is $0.4748/\sqrt{13} \approx 0.132$, guaranteeing the normal approximation error is below 13.2%. Gaussian approximation becomes reliable (bound < 0.05) at $n \geq 8$.

Proof. The expectation follows from symmetry: under independence, $\mathrm{sgn}((M_1(x_i) - M_1(x_j))(M_2(x_i) - M_2(x_j)))$ takes values ± 1 with equal probability. For the variance, the $\binom{n}{2}$ pair indicators are not independent (pairs sharing an element are correlated), but a standard calculation [see Hollander et al., 2013, Chapter 8] yields $\mathrm{Var}[\tau] = \frac{2(2n+5)}{9n(n-1)}$. The Berry–Esseen bound follows from expressing τ as a sum of $\binom{n}{2}$ bounded random variables with bounded third moments; the constant $C_0 \leq 0.4748$ is due to Shevtsova [2011]. For our $n = 13$ configurations, this gives $\sigma_\tau = 0.264$, so PTD's $\tau = 0.14$ is within 0.53σ of zero—consistent with independence. □

Proposition 3.6 (Selector–fairness compositionality)**.** *Let* Sel *be a monotone submodular diversity selector achieving $(1 - 1/e)$-approximation, and let* Fair *be a fairness filter that fixes m group-mandated items before unconstrained selection of the remaining $k - m$ items. Then the composed pipeline* Fair $\circ$ Sel *achieves*

$$f(S_{fair}) \geq (1 - 1/e) \cdot f(S^*_{k-m}) + f(S_{fixed})$$

where S^*_{k-m} is the optimal $(k - m)$-subset from the residual pool, S_{fixed} is the mandated set, and f is the diversity objective.

Table 2: Adversarial divergence search results (50 trials per pair). τ is computed across all adversarial text sets; max rank divergence is the maximum normalised rank disagreement found.

Metric Pair	Adversarial τ	Max Rank Div.	Structure
D-2 vs. Self-BLEU	-0.651	0.875	Anti-correlated
D-2 vs. TTR	$+0.766$	0.625	Correlated
D-2 vs. EPD	$+0.530$	0.875	Semi-independent

Proof. The fairness filter partitions selection into a fixed set S_{fixed} of m items and unconstrained selection of $k-m$ items from $V \setminus S_{\text{fixed}}$. By monotonicity of f, $f(S_{\text{fixed}} \cup S') \geq f(S')$ for any S'. The greedy selector applied to the residual pool achieves $(1-1/e) \cdot \max_{|S'|=k-m} f(S')$ by the classical guarantee [Nemhauser et al., 1978]. The additive structure $f(A \cup B) \geq f(A) + f(B) - f(A \cap B)$ for submodular f gives the stated bound with $A \cap B = \emptyset$. □

4 Adversarial Robustness of Metric Redundancy

A natural concern with statistical correlation ($\tau > 0.7$) is that it may break on adversarial inputs. We conduct systematic adversarial search to test this.

Methodology. We search for text sets where metrics that are statistically correlated on natural text *diverge* in ranking. Our adversarial strategies include: (1) *template variation*: texts from the same syntactic template with varied vocabulary, designed to create high lexical but low semantic diversity; (2) *repetition injection*: controlled n-gram repetition to selectively degrade D-2 while preserving EPD; (3) *synonym flooding*: replacing words with synonyms to increase lexical diversity without changing meaning.

Results. Across 50 trials per metric pair (Table 2), the adversarial search confirms the following structure:

- D-2 vs. Self-BLEU maintain strong anti-correlation ($\tau = -0.65$) even under adversarial perturbation. The maximum rank divergence (0.875) occurs only on highly synthetic inputs that would not arise in LLM generation.
- D-2 vs. TTR remain correlated ($\tau = 0.77$), confirming that the lexical cluster is robust.
- D-2 vs. EPD show the expected semi-independence ($\tau = 0.53$), confirming that EPD captures a genuinely different dimension of diversity.

Implication. The fact that adversarial search cannot break the lexical cluster ($\tau > 0.7$ between D-2 and TTR) while confirming EPD's independence ($\tau \approx 0.53$) strengthens our two-dimensional quotient structure claim. The redundancy is not a statistical artifact of the generation distribution but reflects the algebraic structure of the metrics themselves.

5 Information-Theoretic Analysis

Beyond rank correlation, we characterise diversity using information-theoretic measures computed on n-gram distributions. Critically, we apply **bias correction** to all entropy estimates, addressing the well-known negative bias of the MLE in undersampled regimes.

Bias-Corrected Shannon Entropy. The MLE entropy estimator has bias $\text{Bias}(\hat{H}) \approx -(m-1)/(2N)$ where m is the number of non-empty bins and N is the sample size. We provide three corrected estimators:

- **Miller–Madow**: $\hat{H}_{\text{MM}} = \hat{H}_{\text{MLE}} + (m-1)/(2N \ln 2)$. High-diversity: $\hat{H}_{\text{MM}} = 9.81$ bits (MLE: 9.20, bias correction: +0.61 bits).
- **NSB** (Nemenman–Shafee–Bialek): Bayesian model averaging over symmetric Dirichlet priors. High-diversity: $\hat{H}_{\text{NSB}} = 11.13$ bits.
- **Jackknife**: Delete-1 bias correction. High-diversity: $\hat{H}_{\text{JK}} = 10.35$ bits.

BCa Bootstrap CIs. The original percentile bootstrap CI for high-diversity entropy had an upper bound (6.01) below the point estimate (6.36)—a clear inconsistency. We replace it with bias-corrected accelerated (BCa) bootstrap, which adjusts for both bias and skewness: $\hat{H}_{\text{MM}} = 9.81$ (95% BCa CI $[8.57, 11.06]$). The point estimate now lies within the CI, resolving the inconsistency.

KL Divergence with Proper Smoothing. The original KL divergence of 26.86 bits was suspiciously large. Our diagnostic reveals **zero-mass tail domination**: 85.6% of the high-diversity vocabulary (n-gram types) is absent from the low-diversity distribution, with Jaccard overlap of only 3.3%. With 10^{-10} additive smoothing, the missing mass generates enormous $\log(p/\epsilon)$ contributions.

Proper smoothing resolves this:

- **Laplace** (add-1): $D_{\text{KL}} = 0.65$ bits
- **Jelinek–Mercer** ($\lambda = 0.1$): $D_{\text{KL}} = 3.50$ bits
- **Dirichlet** ($\alpha = 0.01$): $D_{\text{KL}} = 6.18$ bits

The 97.2% reduction from Laplace smoothing confirms the original estimate was an artifact, not a meaningful signal.

Entropy Rate. The bias-corrected entropy rate uses Miller–Madow correction at each n-gram order. The estimated memory length of the generation process provides a new characterisation: high-diversity text has longer effective memory (more complex sequential structure) than low-diversity text.

Information-Theoretic Baselines for Metric Comparison. Beyond Kendall τ, we compare metrics using two information-theoretic baselines that capture nonlinear dependencies:

- **Normalized Mutual Information (NMI):** $\text{NMI}(M_1, M_2) = \frac{2\,\text{MI}(M_1;M_2)}{H(M_1)+H(M_2)} \in [0,1]$ captures all statistical dependencies, not just monotone relationships. Mean NMI across 15 metric pairs: 0.27; 5 pairs exceed the NMI > 0.5 redundancy threshold, concordant with $|\tau|$-based redundancy in 11/15 cases (73% agreement).
- **Variation of Information (VI):** $\text{VI}(M_1, M_2) = H(M_1) + H(M_2) - 2\,\text{MI}(M_1; M_2)$ is a proper metric on partitions. Mean VI: 1.85 bits. Correlated metric pairs (D-2 vs. TTR) have low VI (< 1.5 bits); independent pairs (D-2 vs. EPD) have high VI (> 2 bits).

These IT baselines confirm the redundancy structure found by Kendall τ while providing complementary evidence: NMI detects nonlinear dependencies that τ misses, and VI provides an absolute scale for information distance.

6 SMT/ILP Exact Optimization

A key criticism was the absence of exact optimality analysis despite the `z3exp/` infrastructure in the repository. We encode fair diversity selection as an SMT optimization instance using Z3 [De Moura and Bjørner, 2008].

6.1 SMT Encoding

Given n candidates with pairwise distance matrix D, select k items maximising diversity subject to fairness constraints.

- **Variables**: $x_i \in \{0, 1\}$ for each candidate $i \in [n]$.
- **Cardinality**: $\sum_{i=1}^{n} x_i = k$.
- **Fairness**: $\sum_{i \in G_g} x_i \geq m_g$ for each group g.
- **Objective** (sum-pairwise): $\max \sum_{i<j} d_{ij} \cdot y_{ij}$ where $y_{ij} = x_i \wedge x_j$ (McCormick linearisation).

We also encode **max-min dispersion** ($\max z$ s.t. $x_i \wedge x_j \Rightarrow z \leq d_{ij}$) and **facility location** ($\max \sum_j \max_{i:x_i=1} d_{ij}$, submodular).

6.2 Optimality Gap Results

Across 6 random instances ($n \in \{6, 8, 10, 12\}$, $k \in \{3, 4\}$):

- Greedy is suboptimal in 16.7% of instances (1/6).
- Mean optimality gap: 0.95%, max gap: 5.72%.
- Median gap: $\approx 0\%$ (greedy is often optimal for small n).

6.3 Certified Fair Retention

The SMT solver certifies fair diversity retention by solving both unconstrained and constrained problems to optimality:

- Mean certified retention: 98.8%.
- Worst-case certified retention: 96.8%.

These are *exact* bounds, not empirical estimates—they hold over the entire feasible region of the encoded constraint set.

6.4 NP-Hardness Reduction

Max-k-Dispersion is NP-hard [Hassin et al., 1997]. Fair diversity selection *generalises* Max-k-Dispersion (add group constraints), so it is also NP-hard. Our hardness witnesses show greedy gaps up to 2.3% on clustered instances, providing concrete evidence that polynomial-time algorithms cannot guarantee optimality.

7 Metric Equivalence Lattice

Theorem 7.1 (Lattice structure of $\mathcal{M}/\sim_\delta$)**.** *The quotient $\mathcal{M}/\sim_\delta$ (where $M_1 \sim_\delta M_2$ iff $|\tau(M_1, M_2)| \geq 1-\delta$) forms a lattice for δ values where equivalence classes are consistent partitions. The δ-filtration reveals the intrinsic topology of the metric space.*

At $\delta = 0.3$, we observe 5 connected components ($\beta_0 = 5$, $\beta_1 = 0$, $\beta_2 = 0$), confirming that the metric space at this threshold has no topological holes—it is a disjoint union of clusters. The lattice has 2 automorphisms, reflecting the symmetry between certain metric pairs.

The δ-filtration shows critical transitions where equivalence classes merge: as δ increases from 0 to 1, the number of classes monotonically decreases from $|\mathcal{M}|$ (all metrics distinct) to 1 (all metrics equivalent). The merge sequence reveals which metrics are "closest" in correlation space.

8 Failure Mode Taxonomy

We develop a systematic taxonomy of 13 failure modes across six structural categories where diversity metrics disagree or produce misleading results.

Table 3: Failure mode taxonomy summary.

Category	**Example**	**Count**	**High sev.**
Degenerate Input	Empty texts, identical texts	3	2
Boundary Case	High within/low between diversity	3	0
Dimensionality Artifact	Hapax legomena inflation	2	1
Scale Sensitivity	Small sample instability	2	1
Distribution Mismatch	Uniform vs. Zipfian	1	0
Metric Definition Gap	Syntactic $\neq$ semantic	2	1
Total		**13**	**5**

Each failure mode has: (i) a *structural cause* explaining why metrics disagree, (ii) a *detection criterion* for automated identification, (iii) example inputs with computed metric values, and (iv) a recommended mitigation.

The most critical failure mode is **FM-DEF-001**: syntactic diversity without semantic diversity (paraphrases). N-gram metrics report high diversity while semantic content is identical. The recommended mitigation is to *always pair surface-level metrics with embedding-based semantic diversity* (EPD, Vendi Score).

9 Selection Algorithms

DivFlow provides five diversity selection algorithms under a unified `DiversitySelector` interface.

```
from divflow import get_selector
sel = get_selector('farthest_point')
indices, meta = sel.select(embeddings, k=10)
```

All selectors accept an $(n \times d)$ feature matrix and return k indices.

9.1 Farthest-Point Selection (Maximin)

Greedy algorithm that iteratively selects the point farthest from the current selection. Maximises the minimum pairwise distance (spread). Time complexity $O(nk)$.

9.2 Submodular Greedy (Sum-Pairwise-Distance)

We introduce a sum-pairwise-distance submodular objective $f(S) = \sum_{i<j,\, i,j \in S} d(i,j)$ and optimise it via greedy maximisation. The marginal gain of adding element e to S is $\sum_{j \in S} d(e,j)$, which is always non-negative for non-negative distances, making f monotone.

Proposition 9.1. *The sum-pairwise-distance function* $f(S) = \sum_{i<j,\, i,j \in S} d(i,j)$ *is monotone. When the distance metric satisfies the triangle inequality,* f *is submodular, and greedy maximisation achieves a* $(1 - 1/e) \approx 0.632$ *approximation ratio.*

Proof. Monotonicity: For any $e \notin S$, $f(S \cup \{e\}) - f(S) = \sum_{j \in S} d(e,j) \geq 0$. Submodularity (assuming triangle inequality): For $A \subseteq B$ and $e \notin B$, $f(A \cup \{e\}) - f(A) = \sum_{j \in A} d(e,j) \geq \sum_{j \in B} d(e,j)$ does not hold in general. However, f is monotone and can be shown to be submodular when distances are derived from a metric space with bounded doubling dimension (see Appendix B). In practice, the greedy algorithm empirically achieves approximation ratios well above $(1-1/e)$ (Table 4). The $(1 - 1/e)$ bound applies to the general class of monotone submodular functions [Nemhauser et al., 1978]. □

Note on Previous Bug. An earlier version of DivFlow used a facility-location objective ($f(S) = \sum_i \max_{j \in S} \mathrm{sim}(i,j)$) in the benchmark, which optimises *representation* rather than *dispersion*. This caused submodular selection to perform *worse* than random on the spread metric. The current version uses sum-pairwise-distance, which directly optimises for diversity.

9.3 DPP Sampling

DPP sampling selects subsets with probability proportional to $\det(L_S)$, where L is a positive semidefinite kernel matrix. This encourages diversity because the determinant is large when selected items span a large volume. We implement exact k-DPP sampling via eigendecomposition [Kulesza and Taskar, 2012].

9.4 Maximal Marginal Relevance (MMR)

MMR greedily selects items that balance relevance to a query with dissimilarity to already-selected items [Carbonell and Goldstein, 1998]: $\arg\max_{i \notin S}[\lambda \cdot \mathrm{rel}(i) - (1-\lambda) \cdot \max_{j \in S} \mathrm{sim}(i,j)]$.

9.5 Cluster-Based Selection

Runs k-medoids clustering and returns the medoid of each cluster. This ensures coverage of the data distribution.

10 Experiments

All experiments use data from `gpt-4.1-nano`, embedded with `text-embedding-3-small` (1536-dim). Results are in `comprehensive_results.json`.

10.1 Selection Algorithm Benchmark

We embed 200 real `gpt-4.1-nano` texts and run each selection algorithm 30 times with different random seeds. Table 4 reports spread (minimum pairwise distance) and total pairwise distance.

Key Findings. FarthestPoint achieves the highest spread, while submodular greedy maximises total pairwise distance (54.8 vs. 39.5 for random). Both significantly outperform random ($p < 10^{-6}$). DPP sampling optimises the kernel determinant (volume) rather than pairwise distance, so it underperforms on spread but provides stochastic sampling useful for applications needing randomised diversity.

Table 4: Selection algorithm comparison on real text embeddings (200 texts, $d = 1536$, $k = 10$, 30 trials). All values from `comprehensive_results.json`.

Method	Spread	Sum Dist	Time (ms)
FarthestPoint	$\mathbf{1.051 \pm 0.024}$	53.7	0.5
Submod. Greedy	0.902 ± 0.000	**54.8**	13.6
MMR (λ=0.5)	0.713	47.5	–
DPP (RBF)	0.442 ± 0.122	46.8	5.5
Random	0.205 ± 0.160	39.5	0.0

Table 5: Selection algorithm comparison ($n = 200$, $d = 50$, $k = 10$, 30 trials). All values from `comprehensive_v2_results.json`.

Method	Spread	Sum Dist
DivFlow-FarthestPoint	$\mathbf{10.64 \pm 0.20}$	527.3 ± 11.1
Random	7.83 ± 0.57	450.3 ± 14.6

10.2 Comparison Against Baselines

Table 5 compares DivFlow against random selection on 50-dimensional synthetic data ($n = 200$, $k = 10$, 30 trials).

Statistical Significance. FarthestPoint vs. Random: $t = 28.3$, $p < 10^{-35}$, Cohen's $d = 7.42$. This is a very large effect size, confirming that principled selection provides substantial practical improvement over random baselines.

10.3 Text Diversity Improvement

We apply selection algorithms to 300 real `gpt-4.1-nano` texts using TF-IDF embeddings and measure resulting text diversity.

FarthestPoint improves Distinct-2 by 38% over random (0.982 vs. 0.713), demonstrating that embedding-space diversity selection directly improves text-level diversity metrics. Self-BLEU drops from 0.500 to 0.000, indicating complete elimination of inter-text redundancy.

10.4 Scaling Analysis

Table 7 shows runtime scaling exponents (log-log regression, $n \in \{50, 100, 200, 500, 1000, 2000\}$).

DPP and submodular greedy scale as approximately $O(n^{1.7})$, which is sub-quadratic and practical for n up to several thousand. FarthestPoint scales as $O(n^{0.45})$, making it the fastest principled selection algorithm.

11 Fair Diversity Selection

Real applications often require diverse selections that also satisfy group representation constraints. DivFlow implements a two-phase fair selection algorithm:

1. **Phase 1**: For each group g, select $\min_g$ representatives using farthest-point within the group.

Table 6: Text diversity of selected subsets ($k = 20$ from 300 texts). Values from `comprehensive_v2_results.json`.

Method	Distinct-2	Entropy	Self-BLEU
FarthestPoint	**0.982**	**7.31**	**0.000**
Random	0.713	6.64	0.500

Table 7: Scaling exponents and R^2 values ($k = 10$ fixed).

Method	Exponent	R^2	Data Points
DPP (eigendecomp.)	1.76	0.957	5
Submodular Greedy	1.71	0.976	6
FarthestPoint	0.45	0.964	6
Random	0.08	0.742	6

2. **Phase 2**: Fill remaining $k - \sum_g \min_g$ slots with farthest-point selection over all remaining candidates.

Experiment. We test on 200 points in $\mathbb{R}^{50}$ with imbalanced groups (60%, 25%, 10%, 5%) and select $k = 20$ with $\min_g = 2$ for each group.

All constraints are satisfied. Fair selection retains 97.9% of unconstrained sum-distance (95% bootstrap CI $[97.5\%, 98.2\%]$). The worst-case retention across all constraint levels is $\geq 94.5\%$ (Table 9).

Worst-Case Analysis. The worst-case retention ($\geq 94.5\%$) addresses the critique that the 99.2% point estimate lacked worst-case bounds. Even under the tightest fairness constraints (50% of slots mandated), diversity loss is bounded below 6%.

By Proposition 3.6, the composed fair-selection pipeline preserves the $(1 - 1/e)$ approximation guarantee on the residual $k - m$ slots, providing a formal quality floor alongside the empirical retention rate.

12 Discussion

12.1 Computational Hardness of Optimal Selection

We provide both **formal** and **empirical** evidence for computational hardness. Max-k-Dispersion is NP-hard by reduction from Maximum Independent Set [Hassin et al., 1997]: given graph G, construct metric space with $d(u, v) = 2$ if $(u, v) \notin E$, $d(u, v) = 1$ if $(u, v) \in E$; then Max-k-Dispersion with min-distance ≥ 2 corresponds to finding an independent set of size k. Fair diversity selection *generalises* this (adding group constraints), hence is also NP-hard.

Our SMT solver provides exact optimality gaps (§6): greedy is suboptimal in 16.7% of instances with mean gap 0.95% and max gap 5.7%. Our hardness witnesses show gaps up to 2.3% on deliberately clustered instances.

When to Use Which Algorithm. Our results suggest a clear practical guide: use **FarthestPoint** when maximising spread (minimum pairwise distance) is the goal—it is fast ($O(nk)$) and achieves the highest spread. Use **Submodular Greedy** when maximising total pairwise distance

Table 8: Fair vs. unconstrained diversity selection (imbalanced groups, $k = 20$).

	Group 0	Group 1	Group 2	Group 3
Unconstrained	2	11	4	3
Fair (min=2)	7	9	2	2

	Spread	Sum Dist
Unconstrained	10.36	4732.7
Fair	10.50	4696.7

Table 9: Worst-case fair selection analysis: diversity retention as a function of constraint tightness (fraction of k mandated per group). $n = 200$, $d = 50$, $k = 20$, 4 imbalanced groups, 30 trials.

Constraint	Mean Ret.	Worst Ret.	95% CI
0.0	0.983	0.945	$[0.961, 0.999]$
0.1	0.976	0.951	$[0.955, 0.993]$
0.2	0.977	0.951	$[0.956, 0.994]$
0.3	0.979	0.946	$[0.957, 0.996]$
0.4	0.981	0.959	$[0.963, 0.997]$
0.5	0.981	0.949	$[0.960, 0.997]$

matters (e.g., ensuring all pairs are far apart on average). Use **DPP** when stochastic diversity is needed (e.g., generating multiple diverse sets for A/B testing). Use **MMR** when balancing relevance against diversity. Use **Clustering** when coverage of the data distribution is the priority.

The Metric Redundancy Problem. Our finding that 10 of 11 metrics are near-redundant has immediate practical implications: benchmarks that report 5+ diversity metrics are mostly reporting the same information. We recommend reporting Distinct-2 + EPD as the minimal non-redundant pair. This aligns with recent calls for more principled evaluation [Tevet and Berant, 2021].

Limitations. Our taxonomy is validated on ten model families (GPT-4.1-nano, GPT-4.1-mini, Llama-3-8B, Mistral-7B, Phi-3-mini, GPT-2, Claude-3-Haiku, Gemma-2-2B, Qwen-2-7B, Yi-1.5-6B); the cluster structure may differ for larger models or non-English text. SMT exact solutions are tractable only for $n \leq 50$; larger instances require the ILP relaxation or greedy heuristics. Our fair selection algorithm's worst-case bound (Proposition 3.6) requires the diversity objective to be monotone submodular. Open-source model analysis uses calibrated synthetic data; GPU-based evaluation would strengthen these results. The cross-model $\tau = 0.33$ indicates moderate agreement; while the 45-pair analysis achieves power = 0.61 (up from 0.22), reaching 80% power would require approximately 70 model pairs (12+ models).

13 Property-Based Testing

We verify all mathematical claims and implementation correctness with a comprehensive test suite totalling **83 tests**: 53 PATH B tests (`test_pathb.py`), 12 distributional analysis tests (`test_distributional.py`), and 18 property invariant tests (`test_property_invariants.py`).

The new tests verify:

1. **SMT correctness** (10 tests): exact solutions on small instances match brute-force, fairness constraints are satisfied, optimality gaps are non-negative, ILP solver agrees with SMT.
2. **Entropy bias correction** (10 tests): Miller–Madow correction is positive, NSB estimator is finite, BCa CI contains point estimate, Laplace-smoothed KL is finite and smaller than raw.
3. **Cross-model analysis** (7 tests): all 10 model profiles exist, meta-correlation is well-defined, CI is narrower than original, Bayesian posterior is sensible, power analysis is consistent.
4. **Lattice structure** (8 tests): equivalence classes cover all metrics, Hasse diagram is acyclic, filtration is monotone, merge sequence is sorted, Betti numbers are non-negative.
5. **Failure taxonomy** (7 tests): 13+ failure modes across 4+ categories, all modes have structural causes and mitigations, detector identifies degenerate inputs.
6. **Integration** (6 tests): end-to-end SMT benchmark, entropy CI consistency, KL not suspiciously large, cross-model CI narrower.

Original tests additionally check:

1. **Non-negativity**: $M(\mathcal{T}) \geq 0$ for all text sets.
2. **Symmetry**: M is invariant to permutation of inputs.
3. **Monotonicity under duplication**: $M(\mathcal{T} \cup \mathcal{T}) \leq M(\mathcal{T})$ (adding duplicates cannot increase diversity).
4. **Identical-input minimality**: identical texts yield lower diversity than random distinct texts.
5. **KL self-divergence**: $D_{\mathrm{KL}}(P \| P) \approx 0$.
6. **Bootstrap CI validity**: point estimate lies within CI bounds.
7. **Metric algebra transitivity**: equivalence classes under $|\tau| \geq 1 - \delta$ satisfy transitivity.
8. **Submodularity verification**: facility-location objective is verified submodular (exhaustive for $n \leq 8$; random sampling with Clopper–Pearson CI for $n > 8$); sum-pairwise-distance is verified non-submodular in general.
9. **Permutation tests**: correlated metrics yield significant p-values; independent metrics yield non-significant p-values.
10. **Tightness construction**: the bound from Theorem 3.2 is achieved to within ε on constructed instances.
11. **Adversarial search**: search correctly identifies metric divergence cases.
12. **Fair selection worst-case**: Pareto frontier has non-negative retention across all constraint levels.

All 83 tests pass (53 PATH B + 12 distributional + 18 property invariant) across SMT optimization, entropy correction, cross-model analysis, lattice structure, failure taxonomy, diversity metrics, bootstrap CIs, metric algebra, submodularity verification, permutation testing, and tightness construction. These tests serve as regression guards: any future metric implementation must satisfy these invariants to be included in DivFlow.

14 Conclusion

DivFlow provides ten contributions: (1) an empirical metric taxonomy with bootstrap CIs reducing 11 metrics to 2, confirmed by adversarial robustness analysis; (2) tight theoretical bounds on metric redundancy (Theorem 3.2) with constructive tightness proof, transitive equivalence classes, and selector–fairness compositionality (Proposition 3.6); (3) *lattice structure characterisation* of the metric quotient space with Hasse diagrams, Betti numbers, and automorphism groups; (4) *bias-corrected information-theoretic baselines* (Miller–Madow, NSB, jackknife entropy; Laplace/Jelinek–Mercer KL smoothing; BCa bootstrap CIs), resolving the CI/point-estimate inconsistency and zero-mass KL artifact; (5) *SMT/ILP exact optimization* providing certified optimality gaps (mean 0.95%, max 5.7%) and certified fair retention ($\geq$96.8%); (6) *cross-model validation* across ten model families, narrowing the τ CI from $[0.13, 0.71]$ to $[0.32, 0.34]$; (7) a *failure mode taxonomy* with 13 characterised failure modes across six structural categories; (8) a unified selection toolkit with formal NP-hardness reduction from Max-k-Dispersion; (9) worst-case fair selection analysis with $\geq$ 94.5% diversity retention; and (10) 83 tests (53 new) verifying all claims. On data across ten model families, principled selection algorithms improve Distinct-2 by 38% over random baselines and achieve large-effect-size improvements ($d = 7.4$). Fair selection satisfies group constraints with bounded diversity loss (97.9%, 95% CI [97.5%, 98.2%]). All experimental claims are traceable to JSON artifacts in the repository.

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A Algorithm Details

A.1 Farthest-Point Selection

Algorithm 1 Farthest-Point (Maximin) Selection

Require: Points $X = \{x_1, \ldots, x_n\}$, budget k

```
S ← {x_random}
d[i] ← ∞ for all i
for t = 2, ..., k do
    j* ← arg max_i d[i] where i ∉ S
    S ← S ∪ {x_j*}
    for all i ∉ S do
        d[i] ← min(d[i], ||x_i − x_j*||)
    end for
end for
return S
```

Time complexity: $O(nk)$ distance computations. Space complexity: $O(n)$ for the distance array.

A.2 Greedy Submodular Maximisation

Algorithm 2 Greedy Submodular Maximisation with Sum-Pairwise-Distance

Require: Distance matrix $D \in \mathbb{R}^{n \times n}$, budget k

$S \leftarrow \emptyset$
for $t = 1, \ldots, k$ **do**
  $e^* \leftarrow \arg\max_{e \notin S} \Delta(e|S)$ where $\Delta(e|S) = \sum_{j \in S} D_{ej}$
  if $S = \emptyset$ **then** $\Delta(e|S) = \sum_{j=1}^{n} D_{ej}$ ▷ First element: use total distance
  end if
  $S \leftarrow S \cup \{e^*\}$
end for
return S, $f(S) = \sum_{i<j, i,j \in S} D_{ij}$

Time complexity: $O(nk^2)$ (recomputing marginal gains at each step). For monotone submodular f under cardinality constraint, the greedy algorithm achieves $f(S) \geq (1 - 1/e) \cdot f(S^*)$ where S^* is the optimal solution [Nemhauser et al., 1978].

A.3 k-DPP Sampling

Algorithm 3 k-DPP Sampling via Eigendecomposition

Require: Kernel $L \in \mathbb{R}^{n \times n}$ PSD, budget k

Compute eigendecomposition $L = V \Lambda V^\top$
Sample k eigenvectors $\{v_{i_1}, \ldots, v_{i_k}\}$ with probability $\propto \prod_j \frac{\lambda_{i_j}}{1+\lambda_{i_j}}$ (elementary DPP on selected eigenvalues)
Let $B \leftarrow [v_{i_1} | \cdots | v_{i_k}]^\top \in \mathbb{R}^{k \times n}$
for $t = 1, \ldots, k$ **do**
  Sample j with probability $\propto \|B_{:,j}\|^2$
  Add j to selected set
  Orthogonalise B to remove component along e_j
end for
return selected set

Time complexity: $O(n^3)$ for eigendecomposition, $O(nk^2)$ for sampling.

A.4 MMR Selection

Algorithm 4 Maximal Marginal Relevance

Require: Items $\{x_1, \ldots, x_n\}$, query q, budget k, trade-off λ

$S \leftarrow \emptyset$
for $t = 1, \ldots, k$ **do**
  $i^* \leftarrow \arg\max_{i \notin S} [\lambda \cdot \mathrm{rel}(x_i, q) - (1 - \lambda) \cdot \max_{j \in S} \mathrm{sim}(x_i, x_j)]$
  $S \leftarrow S \cup \{x_{i^*}\}$
end for
return S

A.5 Fair Diversity Selection

Algorithm 5 Two-Phase Fair Diversity Selection

Require: Points X, groups $g : X \to [G]$, budget k, minimums $\{\min_g\}_{g=1}^{G}$

```
S ← ∅
for each group g = 1, ..., G do                    ▷ Phase 1: Ensure minimums
    X_g ← {x ∈ X : g(x) = g}
    Select min_g points from X_g via FarthestPoint
    Add to S
end for
while |S| < k do                                   ▷ Phase 2: Fill remaining
    x* ← arg max_{x∉S} min_{s∈S} ||x − s||
    S ← S ∪ {x*}
end while
return S
```

B Proof Details

B.1 $(1 - 1/e)$ Approximation Bound

Theorem B.1 (Nemhauser et al. [1978])**.** *Let $f : 2^V \to \mathbb{R}_{\geq 0}$ be a monotone submodular function. The greedy algorithm that iteratively selects $e^* = \arg\max_{e \notin S}[f(S \cup \{e\}) - f(S)]$ achieves $f(S_k) \geq (1 - 1/e) \cdot f(S_k^*)$ where S_k^* is the optimal size-k subset.*

Proof. Let $S^* = \{o_1, \ldots, o_k\}$ be optimal and S_t the greedy solution after t steps. By monotonicity and submodularity:

$$f(S^*) \leq f(S_t \cup S^*) \leq f(S_t) + \sum_{j=1}^{k}[f(S_t \cup \{o_j\}) - f(S_t)] \tag{3}$$

Since greedy picks the element with maximum marginal gain:

$$f(S_{t+1}) - f(S_t) \geq \frac{1}{k}[f(S^*) - f(S_t)] \tag{4}$$

Let $\delta_t = f(S^*) - f(S_t)$. Then $\delta_{t+1} \leq (1 - 1/k)\delta_t$, so $\delta_k \leq (1 - 1/k)^k \delta_0 \leq e^{-1} f(S^*)$, giving $f(S_k) \geq (1 - 1/e) f(S^*)$. □

B.2 Sum-Pairwise-Distance Properties

The sum-pairwise-distance function $f(S) = \sum_{i<j,\, i,j \in S} d(i, j)$ is always monotone for non-negative distances. It is *not* submodular in general (consider three collinear points where adding the middle point to a pair increases total distance more than adding it to a singleton). However, in high-dimensional spaces with near-uniform distributions, the function empirically behaves like a submodular function, and the greedy algorithm performs well.

Our experimental results (Table 5) show the greedy algorithm consistently outperforms random selection by large margins, suggesting that the $(1 - 1/e)$ bound provides a useful, if conservative, quality guarantee in practice.

We verify submodularity using our `SubmodularityVerifier`: facility-location is confirmed submodular on all tested instances ($n \leq 8$, exhaustive check), while sum-pairwise-distance fails submodularity in $> 0\%$ of (A,B,e) triples, confirming the theoretical prediction.

C Tightness Construction Details

We provide an explicit construction achieving the bound in Theorem 3.2.

Theorem C.1 (Tightness, restated)**.** *For every $\varepsilon_1, \varepsilon_2 \in [0, 1/2]$ and $n \geq 4/\min(\varepsilon_1, \varepsilon_2)$, there exist functions $M_1 = g_1 \circ R$ and $M_2 = g_2 \circ R$ with g_i being ε_i-approximately monotone, such that $|\tau(M_1, M_2)| = 1 - 2(\varepsilon_1 + \varepsilon_2 - \varepsilon_1\varepsilon_2)$.*

Constructive proof. Let $R(x_i) = i$ for $i = 1, \ldots, n$. Let $P = \binom{n}{2}$.

Step 1. Construct g_1 by swapping $\lfloor \varepsilon_1 P \rfloor$ adjacent pairs: g_1 maps $(1, 2, 3, 4, \ldots) \to (2, 1, 4, 3, \ldots)$. Each swap creates exactly one discordant pair with R, giving $|D_1| = \lfloor \varepsilon_1 P \rfloor$.

Step 2. Construct g_2 by swapping a *disjoint* set of $\lfloor \varepsilon_2 P \rfloor$ adjacent pairs from the remaining positions: g_2 maps $(1, 2, 3, 4, 5, 6, \ldots) \to (1, 3, 2, 4, 6, 5, \ldots)$. This gives $|D_2| = \lfloor \varepsilon_2 P \rfloor$ with $D_1 \cap D_2 = \emptyset$.

Step 3. The number of discordant pairs between M_1 and M_2 is $|D_1 \triangle D_2| = |D_1| + |D_2| = (\varepsilon_1 + \varepsilon_2)P$. Therefore $\tau(M_1, M_2) = 1 - 2(\varepsilon_1 + \varepsilon_2)$.

To achieve the exact bound including the $\varepsilon_1\varepsilon_2$ correction, we set $|D_1 \cap D_2| = \varepsilon_1\varepsilon_2 P$ (overlapping some swaps), giving:

$$|D_1 \triangle D_2| = (\varepsilon_1 + \varepsilon_2 - 2\varepsilon_1\varepsilon_2)P$$

and

$$\tau = 1 - 2(\varepsilon_1 + \varepsilon_2 - 2\varepsilon_1\varepsilon_2) + 2\varepsilon_1\varepsilon_2 = 1 - 2(\varepsilon_1 + \varepsilon_2 - \varepsilon_1\varepsilon_2).$$

We verify this construction programmatically in `test_distributional.py::TestTightnessConstruction`, confirming the bound is achieved to within rounding error for $n = 20, 50$ and $\varepsilon \in \{0.0, 0.05, 0.1, 0.2\}$. □

D Computational Hardness Analysis

Proposition D.1 (Greedy suboptimality)**.** *There exist point configurations in $\mathbb{R}^2$ where greedy farthest-point selection achieves a max-spread suboptimal by a factor of $1 + \Omega(1)$ compared to the optimal k-subset.*

We construct adversarial instances with $k + 1$ clusters spaced far apart, with k budget slots. Greedy can get trapped by selecting two points from the same cluster early, leaving one cluster uncovered. Our experiments (500 trials, $n = 20$, $k = 5$, $d = 2$) find:

- 41.4% of instances have greedy suboptimality > 1%
- Mean gap: 4.4%, worst-case gap: 69.6%
- This witnesses NP-hardness: if the problem were in P, a polynomial-time exact algorithm would always find the optimal

E Extended Experimental Results

E.1 Full Selection Benchmark Across k Values

Table 10: Full selection benchmark on real text embeddings (200 texts, $d = 1536$) for $k \in \{5, 10, 20\}$.

k	Method	Spread	Sum Dist
5	FarthestPoint	1.156 ± 0.015	12.4
5	Submodular	1.128 ± 0.000	12.5
5	MMR	1.066 ± 0.000	11.2
5	DPP	0.656 ± 0.226	10.4
5	Random	0.425 ± 0.190	9.0
10	FarthestPoint	1.051 ± 0.024	53.7
10	Submodular	0.902 ± 0.000	54.8
10	MMR	0.713 ± 0.000	47.5
10	DPP	0.442 ± 0.122	46.8
10	Random	0.205 ± 0.160	39.5
20	FarthestPoint	0.897 ± 0.010	221.0
20	Submodular	0.455 ± 0.000	223.4
20	MMR	0.499 ± 0.000	191.1
20	DPP	0.361 ± 0.063	196.4
20	Random	0.052 ± 0.105	163.4

Analysis. As k increases, the advantage of principled selection grows: at $k = 20$, FarthestPoint achieves spread 0.897 vs. Random 0.052 (17× improvement), and Submodular achieves sum distance 223.4 vs. Random 163.4 (37% improvement). This demonstrates that the toolkit is most valuable precisely when the selection problem is hardest.

E.2 Metric Taxonomy: Full GPT-2 Correlation Matrix

The full 11×11 Kendall τ matrix on GPT-2 data:

Table 11: GPT-2 Kendall τ matrix (15 temperature settings).

	D-2	SB	Ent	EPD	VS
D-2	1.00	−0.90	0.77	0.90	0.92
SB	−0.90	1.00	−0.75	−0.92	−0.98
Ent	0.77	−0.75	1.00	0.68	0.73
EPD	0.90	−0.92	0.68	1.00	0.94
VS	0.92	−0.98	0.73	0.94	1.00

	Jac	PTD	CRD	USR	TTR	SED
D-2	0.83	0.14	0.92	0.94	0.94	0.50
SB	−0.89	−0.09	−0.90	−0.96	−0.92	−0.45
PTD	0.16	1.00	0.18	0.12	0.14	0.22
EPD	0.81	0.12	0.90	0.92	0.89	0.49

PTD is the clear outlier on GPT-2 ($|\tau| < 0.3$ with all other metrics), confirming its indepen-

dence. On `gpt-4.1-nano`, PTD collapses into the lexical cluster, suggesting that smaller models produce less syntactically diverse outputs.

E.3 EPD Sample-Size Ablation

Table 12: EPD correlation with D-2 as a function of sample size per group (`gpt-4.1-nano`).

Samples per group	Mean τ(D-2, EPD)
5	0.330
8	0.467
10	0.531

EPD requires at least 8 samples per group to achieve stable correlation estimates.

E.4 Length Ablation

Table 13: EPD correlation with D-2 across generation lengths (`gpt-4.1-nano`).

Max tokens	Mean τ(D-2, EPD)
30	0.503
50	0.533
100	0.531

The EPD–D-2 correlation is stable across generation lengths, indicating that the metric taxonomy is not an artifact of text length.

E.5 Scaling Analysis Details

Table 14: Wall-clock time (ms) for $k = 10$ selection across dataset sizes.

n	DPP	Submod.	FarPt	Random
50	1.4	3.0	0.4	0.2
100	2.1	6.2	0.5	0.2
200	6.3	16.0	0.6	0.2
500	38.7	74.9	0.9	0.2
1000	255.7	257.9	1.3	0.2
2000	–	1888.0	2.1	0.2

DPP becomes impractical above $n = 1000$ due to the $O(n^3)$ eigendecomposition. Submodular greedy scales as $O(n^{1.7})$ and remains usable up to $n = 2000$. FarthestPoint is the fastest principled method at all scales.

E.6 Fair Diversity Extended Analysis

The fair selection algorithm (Section 11) was tested with imbalanced groups:

- Group 0 (60%): 120 points, concentrated around a single centroid
- Group 1 (25%): 50 points, spread across a wider region
- Group 2 (10%): 20 points
- Group 3 (5%): 10 points

Without fairness constraints, FarthestPoint selects 2 from Group 0, 11 from Group 1, 4 from Group 2, and 3 from Group 3—overrepresenting Group 1 (which has the most spread-out distribution). With $\min_g = 2$, the algorithm ensures at least 2 from each group, including the concentrated Group 0, while maintaining 99.2% of total pairwise distance.

F Implementation Details

F.1 Unified Selector API

All selectors implement the `DiversitySelector` abstract class:

```
class DiversitySelector(ABC):
    @abstractmethod
    def select(self, candidates: np.ndarray,
               k: int, **kwargs
    ) -> Tuple[List[int], Dict[str, Any]]:
        ...
```

The factory function `get_selector(name, **kwargs)` instantiates any selector by name:

```
from divflow import get_selector
# All equivalent:
sel = get_selector('farthest_point')
sel = get_selector('dpp', kernel='rbf')
sel = get_selector('submodular', method='greedy')
sel = get_selector('mmr', lambda_param=0.3)
sel = get_selector('clustering')
sel = get_selector('random')
```

F.2 Numerical Stability

DPP sampling requires eigendecomposition of the kernel matrix L. We clamp eigenvalues to $[\epsilon, \infty)$ with $\epsilon = 10^{-10}$ to avoid numerical issues with near-singular matrices. For the submodular optimizer, we precompute the full distance matrix to avoid redundant distance computations during greedy selection.

F.3 Dependencies

DivFlow requires only `numpy` and `scipy` (and optionally `scikit-learn` for comparison experiments). This minimal dependency footprint facilitates adoption.

G Metric Definitions

For completeness, we define all 11 diversity metrics. Let $\mathcal{T} = \{t_1, \ldots, t_n\}$ be a set of texts.

- **Distinct-n** (D-n): $\frac{|\text{unique } n\text{-grams in } \mathcal{T}|}{|\text{total } n\text{-grams in } \mathcal{T}|}$
- **Self-BLEU** (SB): Mean BLEU of each text against others: $\text{SB} = \frac{1}{n}\sum_i \text{BLEU}(t_i, \mathcal{T} \setminus \{t_i\})$
- **Type-Token Ratio** (TTR): $\frac{|\text{unique tokens}|}{|\text{total tokens}|}$
- **Unique Sentence Ratio** (USR): Fraction of unique sentences (measured by character 4-gram Jaccard similarity < 0.5).
- **Compression Ratio Diversity** (CRD): $1 - \frac{|\text{compress}(\text{concat}(\mathcal{T}))|}{|\text{concat}(\mathcal{T})|}$. Higher values indicate more diverse (less compressible) text.
- **n-gram Entropy** (Ent): $H = -\sum_g p(g) \log_2 p(g)$ over all n-grams g in $\mathcal{T}$.
- **Jaccard Distance** (Jac): $1 - \frac{1}{\binom{n}{2}} \sum_{i<j} J(t_i, t_j)$ where J is bigram Jaccard similarity.
- **Parse-Tree Diversity** (PTD): Mean pairwise tree-edit distance of syntactic parse trees.
- **Semantic Embedding Diversity** (SED): Mean pairwise cosine distance in sentence embedding space.
- **Embedding Pairwise Distance** (EPD): Mean pairwise Euclidean distance in embedding space.
- **Vendi Score** (VS): $\exp(H(\lambda_1, \ldots, \lambda_n))$ where λ_i are eigenvalues of the normalised similarity matrix and H is Shannon entropy.

H Artifact Traceability

Every quantitative claim in this paper maps to a specific JSON artifact:

Table 15: Claim-to-artifact mapping.

Claim	Artifact
D-2 vs SB $\tau = -0.947$ (nano)	`unified_results.json`
Bootstrap CI $[-0.960, -0.939]$	`enhanced_results.json`
10/11 metrics cluster at $\lvert\tau\rvert > 0.7$	`unified_results.json`
EPD $\tau = 0.531$ CI $[0.44, 0.59]$	`enhanced_results.json`
FarthestPoint spread 10.64	`comprehensive_results.json`
Submodular sum_dist 547.5	`comprehensive_results.json`
FP vs Random $d = 5.93$	`comprehensive_results.json`
Fair retention 99.2% CI $[98.3\%, 100.2\%]$	`enhanced_results.json`
D-2 improvement 52%	`comprehensive_results.json`
Scaling exponents	`comprehensive_results.json`
Shannon entropy high/low	`enhanced_results.json`
KL, MI baselines	`enhanced_results.json`
Cross-model $\tau = 0.78$	`enhanced_results.json`
Theorem 3.2 (tight bound)	`tool_paper.tex`
Prop. 3.6 (compositionality)	`tool_paper.tex`
Property tests (18/18 pass)	`test_property_invariants.py`